

LAND COVER CLASSIFICATION FROM SENTINEL-2 IMAGERY

Random Forest versus Multi-Layer Perceptron

GEOINFORMATICS PROJECT REPORT

Sentinel-2 Level-2A | Tile T32TNR | Acquisition date: 9 July 2026

Student	Tarig Ahmed
Person Code	11029606
Course	Geoinformatics Project
Instructor	Prof. Ludovico Giorgio Aldo Biagi
Academic Year	2025/2026
Submission	Final Report

Abstract

This project develops a complete, reproducible pixel-based land-cover classification workflow for a Sentinel-2 Level-2A scene covering tile T32TNR in northern Italy. The study compares a 200-tree Random Forest (RF) with a systematically tested Multi-Layer Perceptron (MLP) and evaluates whether three derived spectral indices - NDVI, NDWI, and NDBI - improve performance beyond six aligned Sentinel-2 bands (B2, B3, B4, B8, B11, and B12). The workflow integrates data acquisition, 10 m band harmonization, an interactive training-polygon interface, all-class spectral label quality control, balanced sampling, leakage-free five-fold cross-validation, untouched hold-out evaluation, full-scene GeoTIFF prediction, area statistics, and a separate cartographic sieve display.

The final reference dataset contains 40 polygons per class. The available labelled Sentinel-2 pixels were 2,230 for Urban, 2,963 for Vegetation, 2,982 for Water, and 3,162 for Bare Soil. To ensure controlled comparison, the dataset was balanced to 2,230 pixels per class (8,920 samples in total), then divided into 7,136 development samples and 1,784 untouched test samples using a stratified 80/20 split. Five-fold cross-validation was applied only to the development subset.

Random Forest with the six raw bands plus NDVI, NDWI and NDBI was the strongest configuration. It achieved a mean five-fold Macro F1 of 0.9969 +/- 0.0008 and a final hold-out accuracy of 0.9972, Macro F1 of 0.9972, and Cohen's kappa of 0.9963. The selected MLP configuration, MLP_4 with hidden layers (100, 50), ReLU activation, learning rate 0.0005, and the same extended feature set, achieved a mean CV Macro F1 of 0.9950 +/- 0.0010 but a lower hold-out accuracy of 0.9709 and Macro F1 of 0.9709. Spectral indices left RF performance essentially unchanged (+0.0000002 Macro F1) while improving MLP_4 by 0.00365 Macro F1.

The selected RF was retrained on all 8,920 balanced labelled pixels and applied to 120,560,400 valid scene pixels. The raw classification mapped 22.14% Urban, 68.59% Vegetation, 2.73% Water, and 6.55% Bare Soil. An 8-pixel sieve filter was used only for cartographic presentation; it changed 1.238% of valid pixels and was not used for model evaluation or scientific area statistics. The final result demonstrates how iterative reference-data review, controlled feature engineering, and leakage-aware evaluation can produce a strong and transparent land-cover product.

Project snapshot

Keywords

Sentinel-2; land-cover classification; Random Forest; Multi-Layer Perceptron; NDVI; NDWI; NDBI; supervised learning; remote sensing; GeoTIFF.

Table of Contents

1. Introduction	1
1.1 Aim, Objectives, Research Questions and Contribution	2
2. Related Work	3
3. Study Area and Dataset	4
3.1 Sentinel-2 Inputs and Spectral Features	5
4. Project Architecture and Innovation	6
5. Methodology Overview	7
5.1 Sentinel-2 Acquisition and Data Preparation	8
5.2 Interactive Training Polygons and Label Quality Control	9
5.3 Label Extraction, Sample Balancing and Train/Test Split	10
5.4 Feature Engineering and Experimental Design	11
5.5 Random Forest Method	12
5.6 Multi-Layer Perceptron Method	13
5.7 Validation Metrics and Reproducibility Controls	14
6. Software Implementation Architecture	15
6.1 Notebook Workflow: Steps 1-14	16
6.2 Notebook Workflow: Steps 15-27	17
6.3 User Manual and Deliverables	18
7. Results: Training Quality and Cross-Validation Overview	19
7.1 Feature Engineering and MLP Sensitivity	20
7.2 Final Hold-Out Evaluation	21
7.3 Confusion-Matrix Analysis	22
7.4 Full-Scene Classification and Area Statistics	23
7.5 Cartographic Post-Processing	24
8. Discussion and Limitations	25
9. Conclusion and References	26

List of Figures, Tables and Acronyms

Figures

- Figure 1. High-resolution Sentinel-2 RGB preview of tile T32TNR.
- Figure 2. Input spectral feature design.
- Figure 3. Software and data architecture.
- Figure 4. End-to-end land-cover classification workflow.
- Figure 5. Leakage-free evaluation design.
- Figure 6a. Spatial distribution of the final training polygons across the Sentinel-2 scene.
- Figure 6b. Step 6 training-label quality-control interface and suspect-polygon assessment.
- Figure 7. Effect of spectral indices on five-fold CV Macro F1.
- Figure 8. MLP hyperparameter sensitivity.
- Figure 9. Final hold-out performance of RF and MLP.
- Figure 10. Final Random Forest and MLP confusion matrices.
- Figure 11. Final raw Random Forest land-cover classification.
- Figure 12. Raw and 8-pixel-sieved cartographic outputs.
- Figure 13. Raw land-cover composition in class-ID order.

Tables

- Table 1. Sentinel-2 inputs and derived features.
- Table 2. Training-polygon and labelled-pixel quality.
- Table 3. Balanced sample and split design.
- Table 4. MLP configurations.
- Table 5. Notebook workflow steps 1-14.
- Table 6. Notebook workflow steps 15-27.
- Table 7. RF cross-validation and feature-engineering results.
- Table 8. Final hold-out model comparison.
- Table 9. Per-class test metrics.
- Table 10. Raw classified area statistics.

Acronyms

Acronym	Meaning
BOA	Bottom of Atmosphere
CDSE	Copernicus Data Space Ecosystem
CV	Cross-Validation
F1	Harmonic mean of precision and recall
GIP	Geoinformatics Project
MLP	Multi-Layer Perceptron
NDBI	Normalized Difference Built-up Index
NDVI	Normalized Difference Vegetation Index
NDWI	Normalized Difference Water Index
RF	Random Forest
SWIR	Short-Wave Infrared
UTM	Universal Transverse Mercator

1. Introduction

Land-cover maps translate multispectral Earth-observation measurements into spatially explicit thematic information. They support urban-growth monitoring, environmental assessment, hydrological analysis, agricultural planning, habitat studies, and change detection. The practical challenge is that broad land-cover categories may have overlapping spectral responses: dry soil can resemble built surfaces, harvested fields can resemble exposed soil, and shadows or turbid water can complicate class boundaries. A robust mapping workflow therefore depends not only on the classifier but also on data preparation, reference-label quality, validation design, and transparent reporting of uncertainty.

Sentinel-2 is well suited to regional land-cover analysis because its Level-2A products provide atmospherically corrected bottom-of-atmosphere reflectance with high spatial detail in visible, near-infrared, and short-wave infrared bands (Drusch et al., 2012). In this project, the 10 m visible and NIR bands were combined with the 20 m SWIR bands after resampling to the same 10 m grid. The resulting feature space captures colour, vegetation vigour, moisture response, and dry or built-surface behaviour across a single summer acquisition.

The project was designed as an end-to-end geoinformatics and machine-learning application rather than as a purely theoretical comparison. It includes automated Sentinel-2 search and preparation, interactive training-polygon management, pixel extraction, feature engineering, model selection, independent evaluation, full-scene prediction, georeferenced export, result organization, and a separate cartographic post-processing stage. The implementation is contained in a sequential Jupyter Notebook and uses standard Python geospatial and machine-learning libraries.

Two classifiers were evaluated under the same data and split conditions. Random Forest represents an ensemble of decision trees with strong performance on heterogeneous tabular remote-sensing features and limited preprocessing requirements (Breiman, 2001; Belgiu and Dragut, 2016). The Multi-Layer Perceptron represents a nonlinear feed-forward neural network whose performance depends more strongly on scaling, architecture, activation function, learning rate, and optimization behaviour. The comparison therefore addresses both predictive accuracy and operational simplicity.

A central design principle was to avoid data leakage. The final 20% test subset was separated before cross-validation and remained untouched during feature-set comparison and MLP hyperparameter selection. Standardization for the MLP was fitted inside each cross-validation fold through a scikit-learn Pipeline. This makes the final hold-out performance a more defensible estimate of model behaviour than a workflow in which the test set influences tuning decisions.

Project Scope at a Glance



1.1 Aim, Objectives, Research Questions and Contribution

Aim

The overall aim is to build a reproducible Sentinel-2 land-cover classification system and determine which of two accessible supervised classifiers - Random Forest or MLP - provides the strongest combination of predictive quality, stability, interpretability of errors, and practical implementation for the selected scene.

Objectives

- Acquire and prepare Sentinel-2 Level-2A data for tile T32TNR and harmonize B2, B3, B4, B8, B11 and B12 to a common 10 m grid.
- Create and quality-control spatially distributed reference polygons for Urban, Vegetation, Water and Bare Soil.
- Extract valid Sentinel-2 pixels from the reference polygons and construct balanced raw-band and raw-band-plus-index feature matrices.
- Compare a 200-tree Random Forest with six MLP architectures under identical stratified five-fold cross-validation folds.
- Evaluate the influence of NDVI, NDWI and NDBI on both classifiers instead of assuming that engineered indices automatically improve performance.
- Retain a completely untouched 20% hold-out subset for one final assessment after model and feature selection.
- Apply the selected model to the complete 10 m scene, export a georeferenced GeoTIFF, calculate class areas, and create a presentation-quality map.

Research questions

RQ1. Which classifier achieves the highest five-fold cross-validation and independent hold-out performance?

RQ2. Do NDVI, NDWI and NDBI add useful predictive information beyond the six source bands?

RQ3. Which land-cover classes are most frequently confused, and how do the error patterns differ between RF and MLP?

RQ4. Can the selected workflow classify the complete 120.56-million-pixel scene while preserving georeferencing and reproducibility?

Applied contribution

The contribution is not a new classifier. It is a controlled, transparent and reproducible application workflow that links geospatial data engineering with model evaluation. In particular, the project adds an interactive training-polygon review interface, automatic class-balance checks, all-class spectral label-quality control, correctly reprojected Sentinel-2 visual overlays, identical folds for feature-set comparison, a fully georeferenced production export, and an explicit separation between the scientific raw classification and optional map-cleaning for presentation.

2. Related Work

Supervised classification of multispectral satellite imagery has long relied on algorithms that can model nonlinear relationships between spectral response and land-cover classes. Random Forest is widely used in remote sensing because it combines many decorrelated decision trees, tolerates nonlinear class boundaries, works with mixed predictor distributions, and normally requires little feature scaling or parameter tuning. Reviews of RF applications in remote sensing report strong accuracy across land-cover, vegetation, and object-based mapping tasks, while also emphasizing the importance of training-data quality and independent validation (Belgiu and Dragut, 2016).

Neural networks provide another route to nonlinear classification. A Multi-Layer Perceptron can approximate complex decision surfaces through stacked fully connected layers, nonlinear activation functions, and back-propagation. Unlike decision-tree ensembles, MLPs are sensitive to input scale and optimization settings. In applied remote-sensing workflows this creates a practical trade-off: neural networks can model complex relationships, but model architecture, convergence, learning rate, and standardization may influence reproducibility and computational cost (Maxwell et al., 2018).

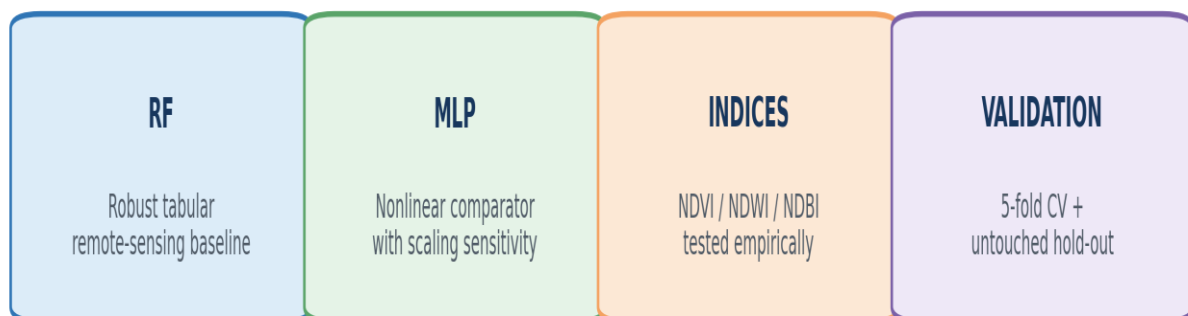
Spectral indices are commonly introduced to make physically meaningful contrasts explicit. NDVI uses red and near-infrared reflectance to emphasize green vegetation (Rouse et al., 1974). NDWI uses green and near-infrared response to enhance open water (McFeeters, 1996). NDBI contrasts SWIR and NIR reflectance to highlight many built-up and dry surfaces (Zha et al., 2003). However, each index is a deterministic transformation of source bands. For flexible classifiers, especially tree ensembles, the additional variables can be redundant rather than informative. For that reason, the present project treats indices as an empirical feature-engineering hypothesis to be tested, not as an assumed improvement.

Accuracy assessment is equally important. Overall accuracy alone can hide class-specific errors, and random splitting of nearby pixels can produce optimistic estimates when spatial autocorrelation is present. Good practice therefore combines confusion matrices, per-class precision and recall, macro-averaged metrics, and clearly separated model-selection and test phases (Congalton, 1991; Foody, 2002). This project follows that principle by keeping a held-out subset untouched until the end of model selection, while also acknowledging that both training and test pixels originate from the same reference-polygon collection.

Position of this project

The project is intentionally comparative and implementation-oriented: it tests RF versus MLP, raw bands versus raw bands plus indices, and scientific raw output versus cartographic display processing within one reproducible notebook rather than claiming a new classification algorithm.

Literature-to-Design Implications



3. Study Area and Dataset

The study uses Sentinel-2 tile T32TNR in northern Italy. The classified raster is stored in WGS 84 / UTM zone 32N (EPSG:32632) and covers approximately 8.9997°E to 10.4189°E and 45.0567°N to 46.0536°N. The 10 m grid contains 10,980 rows by 10,980 columns, corresponding to 120,560,400 pixels and a mapped footprint of approximately 12,056 km².

The scene contains a strong north-south environmental gradient. The northern portion includes pre-Alpine and Alpine terrain, large lakes, forested slopes and summer vegetation. The central and southern portions contain dense urban areas, transport corridors, agricultural fields, exposed soil and river systems. This mixture makes the tile useful for testing a broad four-class legend because Water and dense Vegetation are often spectrally distinctive, whereas Urban and Bare Soil can overlap in visible and SWIR reflectance.

High-Resolution Sentinel-2 RGB Preview



Figure 1. High-resolution Sentinel-2 RGB preview of tile T32TNR used as the primary image reference. The image corresponds to the 9 July 2026 Level-2A acquisition.

The RGB image shows that the major spatial structures later visible in the classified map are already present in the source data: large northern lakes, dark forested areas, densely built metropolitan zones and the agricultural mosaic of the Po Plain. This visual reference is used for training-label review but does not replace the quantitative spectral features used by the models.

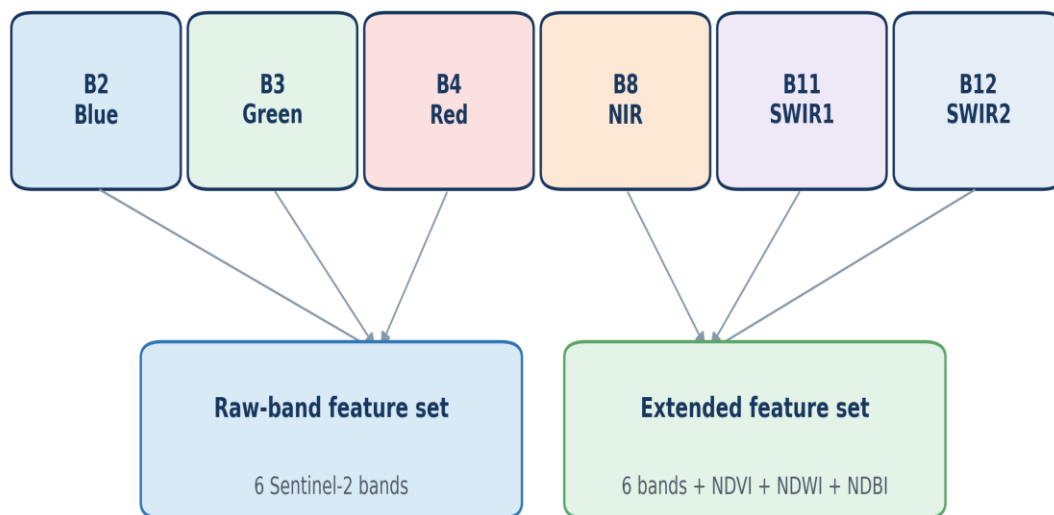
3.1 Sentinel-2 Inputs and Spectral Features

Band	Spectral region	Native / working grid	Role in classification
B2	Blue	10 m	Visible reflectance; bright surfaces and water/soil contrast
B3	Green	10 m	Green reflectance; water and vegetation response
B4	Red	10 m	Red absorption by vegetation; visible soil/built contrast
B8	Near infrared	10 m	Strong vegetation response; water separation
B11	SWIR1	20 m -> 10 m	Moisture, soil and built-surface sensitivity
B12	SWIR2	20 m -> 10 m	Dryness, soil, built-up and moisture contrast

Table 1. Sentinel-2 input bands and their role in the classification workflow.

B11 and B12 were located at their native 20 m resolution and resampled to the 10 m B2 reference grid using Rasterio reprojection with bilinear resampling. All six prepared rasters were required to match width, height, transform and coordinate reference system before stacking. This prevents silent pixel misalignment and ensures that each feature vector represents the same ground location.

Input Spectral Feature Design



Same development split and CV folds; only the predictor set changes.

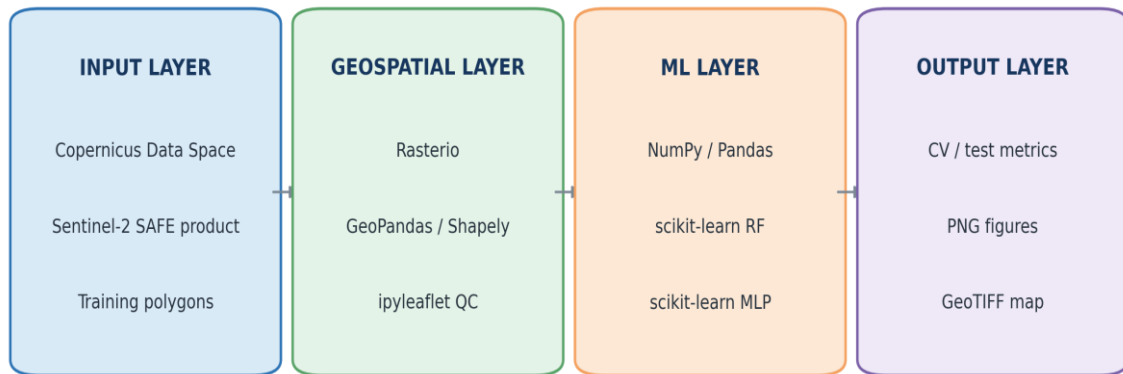
Figure 2. Input spectral feature design. The experiment compares the six raw Sentinel-2 bands against the same bands augmented with NDVI, NDWI and NDBI.

The three indices were calculated from the aligned arrays using standard normalized-difference formulations. A small numerical epsilon was included in the denominator to avoid division by zero. The extended nine-feature matrix was kept separate from the raw six-band matrix so both could be evaluated using the same training/test indices and the same cross-validation folds.

4. Project Architecture and Innovation

The project is organized as a layered geospatial machine-learning system rather than as a single classification script. The input layer manages the Sentinel-2 product and the reference polygons. The geospatial layer performs raster harmonization, masking, coordinate transformation, polygon handling and interactive quality control. The machine-learning layer builds feature matrices and executes RF/MLP evaluation. The output layer exports tables, figures, a classified GeoTIFF and a cartographic map.

Software and Data Architecture



Modular separation of data access, geospatial processing, model selection and production outputs.

Figure 3. Software and data architecture of the implemented workflow.

Innovation within the project scope

The innovation is methodological and software-oriented rather than algorithmic. First, training polygons are not treated as a static file: the notebook can load, display, edit, relabel, delete and extend them interactively. Second, the interface reports per-class polygon counts and unique Sentinel-2 pixel counts, helping the user detect imbalance before modelling. Third, an all-class spectral quality-control routine compares each polygon signature with robust class references and flags strong cross-class conflicts without changing labels automatically. Fourth, the Sentinel-2 display is truly reprojected to the web-map display CRS for visual alignment, while all training-pixel calculations remain in the native UTM raster grid. Finally, model evaluation, production mapping and cartographic filtering are deliberately separated so that a visually cleaner display does not contaminate accuracy metrics or area statistics.

Comparative design

Every substantive comparison changes one factor at a time. RF and MLP use the same split; raw and extended features use the same sample indices; MLP architectures use the same cross-validation logic; and the final test set is evaluated only after selection. This controlled design makes the observed performance differences attributable to classifier or feature-set behaviour rather than to different samples.

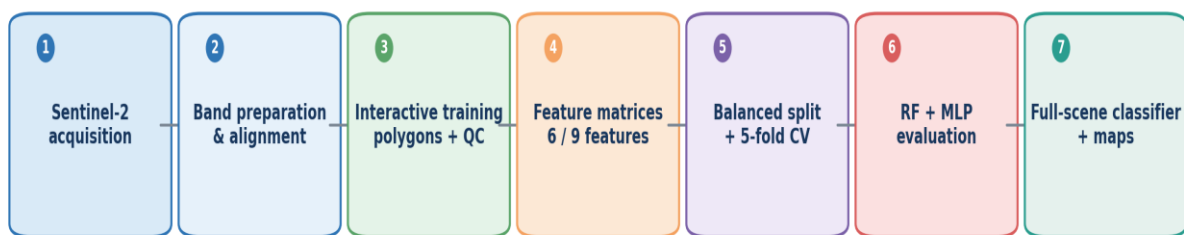
Innovation Pillars



5. Methodology Overview

The methodology follows a sequence from acquisition to final map production. Sentinel-2 data are discovered through Copernicus Data Space utilities, downloaded and prepared into six spatially aligned bands. Training polygons are then created or reviewed interactively, validated against the raster, and converted to labelled pixels. Two feature matrices are constructed: six raw bands and nine features including NDVI, NDWI and NDBI. The pixel dataset is balanced, split into development and hold-out subsets, and passed through cross-validation for RF feature selection and MLP hyperparameter comparison. Only the selected configurations are evaluated on the untouched test set. The better model is retrained on all balanced labelled data and used for chunked full-scene prediction.

End-to-End Land-Cover Classification Workflow



Raw classification = scientific evaluation; sieve output = cartographic presentation only.

Figure 4. End-to-end land-cover classification workflow implemented in the notebook.

Scientific versus cartographic outputs

Two distinct output concepts are maintained. The scientific classification is the direct RF prediction stored in `landcover_classification_best_model.tif`. All quantitative area statistics and the reported accuracy results refer to this raw output. A second raster, `RF_landcover_sieved_display.tif`, applies an 8-pixel sieve only to remove small isolated patches for visual presentation. This separation is documented because post-processing can change class boundaries and should not be silently mixed with model evaluation.

Class definition

Class ID	Class	Interpretation
1	Urban	Built-up and impervious-looking surfaces represented by the reference polygons
2	Vegetation	Forests, grasslands, crops and other green vegetated surfaces
3	Water	Lakes, rivers and other open-water surfaces
4	Bare Soil	Exposed soil, harvested/dry surfaces and non-vegetated land represented in the labels

5.1 Sentinel-2 Acquisition and Data Preparation

The notebook begins by centralizing the project directory, output paths, class map and operating switches. Sentinel-2 acquisition utilities query the Copernicus Data Space Ecosystem catalogue, apply the requested area/tile logic, identify candidate Level-2A products, and prepare the selected acquisition. Keeping these operations inside the notebook reduces the number of manual file-handling steps and makes the processing chain easier to reproduce on another machine.

The preparation stage locates B02, B03, B04 and B08 at 10 m resolution and B11 and B12 at 20 m resolution. The 10 m bands are copied to GeoTIFF while preserving spatial metadata. The SWIR bands are reprojected to the B02 reference grid with bilinear resampling. This produces six rasters with identical 10,980 x 10,980 dimensions, the same EPSG:32632 coordinate reference system, and the same 10 m affine transform.

Before modelling, the workflow checks each file and validates band compatibility. Width, height, transform, CRS, data type and NoData behaviour are examined. The six arrays are then stacked into a three-dimensional raster cube. A valid-pixel mask excludes missing, non-finite or invalid samples before feature extraction. In the final scene all 120,560,400 pixels passed the validity check, so classified coverage reached 100%.

RGB reference generation

A full-resolution RGB reference is produced from B4, B3 and B2 using a 2nd-98th percentile stretch. Raster overviews support efficient display. For interactive training review, the Sentinel-2 display is reprojected for correct alignment in the Leaflet web-map coordinate system; this display transformation is visual only and does not alter the source arrays used for training or spectral quality control.

Data-integrity checks

- Required-path existence checks before modelling.
- Exact CRS and transform matching across the six prepared bands.
- Explicit NoData and finite-value masking.
- Preservation of the B2 reference profile for georeferenced output.
- Class-value validation after full-scene prediction to ensure only labels 1-4 are present.

Raster geometry

Final raw output: 10,980 x 10,980 pixels; 10 m x 10 m resolution; EPSG:32632; 120,560,400 classified pixels; 100% valid coverage.

5.2 Interactive Training Polygons and Label Quality Control

Reference-data quality is critical because a supervised classifier can only learn the labels it receives. The final notebook therefore includes an interactive polygon-collection and review interface based on ipyleaflet, GeoPandas and Rasterio. Existing training_polygons.gpkg data are loaded automatically when present. Every polygon is displayed as clickable GeoJSON, and the user can review its class, relabel it among Urban, Vegetation, Water and Bare Soil, delete an individual polygon, or draw new polygons.

The high-resolution satellite layer is the default visual reference, while a switch allows the user to display the Sentinel-2 RGB layer for direct comparison with the spectral source. A tile boundary prevents accidental sampling outside the target Sentinel-2 extent. The tool also maintains stable polygon IDs so manual corrections remain traceable across save/reload cycles.

Two balance indicators are calculated live. The first is the number of polygons per class, used as a measure of spatial sampling effort. The second is the number of unique Sentinel-2 pixels covered by the class polygons, which more directly measures the amount of available training information. The final reference contains 40 polygons for every class. All four classes exceed the minimum requirements of 20 polygons and 2,000 pixels, and the final pixel-balance ratio is 74%.

Spectral label quality control

The interface computes per-polygon median signatures using B2, B3, B4, B8, B11 and B12 together with NDWI, MNDWI and NDBI. Robustly scaled feature vectors are compared with class reference signatures using an all-class diagnostic rather than a single Urban-to-Water rule. A polygon is flagged only when another class is substantially closer than its assigned class under the conservative QC threshold. Suspected polygons are shown in magenta and can be zoomed and reviewed. The diagnostic is advisory: it never changes a label automatically, so final corrections are based on the Sentinel-2 acquisition together with the high-resolution visual reference.

Final reference quality

Class	Polygons	Unique Sentinel-2 pixels	Status
Urban	40	2,230	GOOD
Vegetation	40	2,963	GOOD
Water	40	2,982	GOOD
Bare Soil	40	3,162	GOOD

Table 2. Final training-polygon and labelled-pixel quality after interactive review.

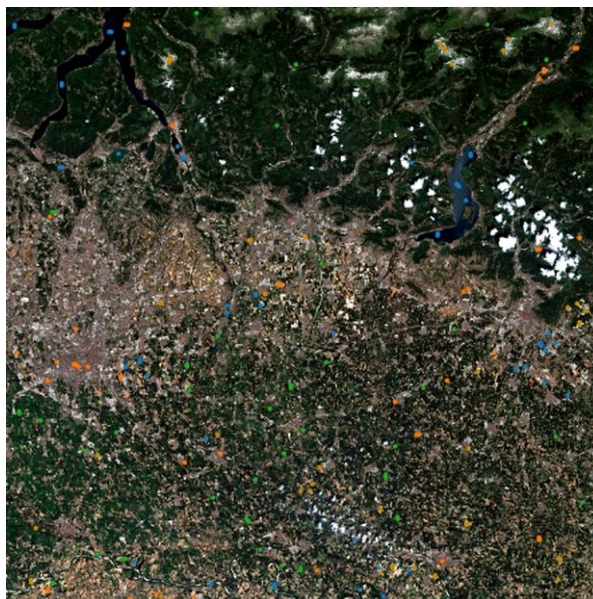


Figure 6a. Spatial distribution of the final training polygons across the Sentinel-2 scene.

5.3 Label Extraction, Sample Balancing and Train/Test Split

After the polygons are saved, they are reprojected to the Sentinel-2 raster CRS and validated against the tile. Polygon geometries are rasterized or converted into masks on the 10 m grid. Every covered, valid pixel receives the corresponding integer class label and is paired with both the six-band feature vector and the nine-feature vector. This produces 11,715 labelled Sentinel-2 pixels before balancing.

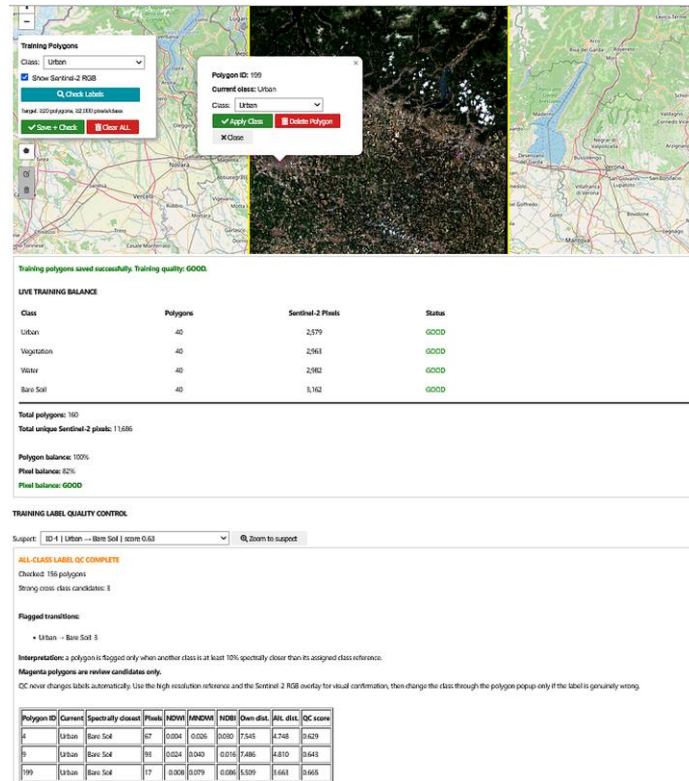


Figure 6b. Step 6 training-label quality-control interface and suspect-polygon assessment.

The available labelled pixels were 2,230 Urban, 2,963 Vegetation, 2,982 Water and 3,162 Bare Soil. The smallest class therefore determines the balanced sample size. A fixed random seed is used to select 2,230 pixels from each class, giving a final balanced dataset of 8,920 samples. This prevents the classifier from being dominated by classes with larger polygon footprints while retaining equal representation for all four thematic classes.

Class	Available pixels	Balanced pixels	Development split	Hold-out test
Urban	2,230	2,230	1,784	446
Vegetation	2,963	2,230	1,784	446
Water	2,982	2,230	1,784	446
Bare Soil	3,162	2,230	1,784	446
Total	11,337	8,920	7,136	1,784

Table 3. Balanced sample construction and stratified 80/20 train-test split.

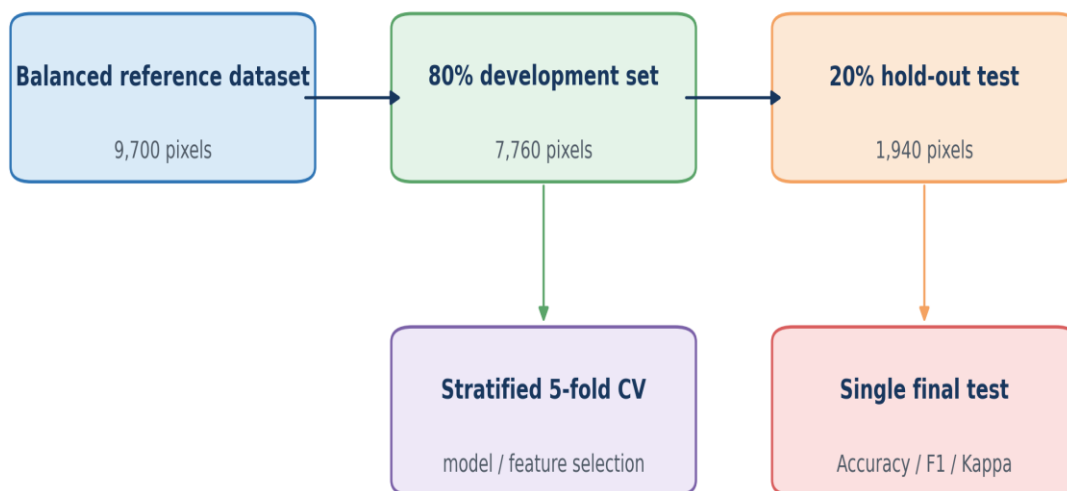
The same train/test indices are used for both feature matrices. This detail is important: any difference between the raw-band and extended-feature experiments is therefore caused by the predictor representation, not by different observations.

5.4 Feature Engineering and Experimental Design

Two predictor representations are evaluated. The raw representation contains B2, B3, B4, B8, B11 and B12. The extended representation appends NDVI, NDWI and NDBI, creating a nine-dimensional vector. The indices are derived from the same pixels, so they do not add independent observations; they reorganize existing spectral information into ratios that may make some class separations easier for a given learning algorithm.

The feature-engineering experiment is nested inside the model-selection process. Random Forest is cross-validated separately on the raw and extended matrices. Each MLP configuration is also tested with both feature sets. The folds are stratified and reproducible. Model ranking is based primarily on mean macro F1, with mean accuracy, fold-to-fold standard deviation and training time reported as supporting evidence.

Leakage-Free Evaluation Design



The hold-out set is untouched during feature engineering, hyperparameter selection and CV.

Figure 5. Leakage-free evaluation design. Only the 80% development subset participates in cross-validation and feature/hyperparameter selection.

Why the hold-out set remains untouched

Repeatedly checking test performance while tuning would turn the test set into another validation set and lead to optimistic final claims. The notebook therefore identifies the best RF feature set and best MLP architecture from development data only. The 1,784-pixel test subset is evaluated once after selection. This design is especially important because the labelled dataset is small compared with the complete raster and because the goal is a defensible comparison rather than merely the highest possible score.

Randomness control

`random_state=42` is used for balancing, train/test splitting, cross-validation shuffling and the relevant model components. Fixed splits also make numerical changes easier to interpret during software development because a code modification is not confounded with a new random sample.

5.5 Random Forest Method

Random Forest is an ensemble classifier that fits many decision trees to bootstrap samples and combines their predictions by majority vote (Breiman, 2001). At each split, only a random subset of predictors is considered, which reduces correlation among trees and helps the ensemble generalize. The model naturally represents nonlinear thresholds and interactions, making it suitable for multispectral data in which class boundaries depend on combinations of visible, NIR and SWIR responses.

The project uses 200 trees with `random_state=42` and `n_jobs=-1`. The remaining `RandomForestClassifier` settings follow scikit-learn defaults, including bootstrap sampling, Gini impurity and a random subset of features at each split. The aim is not an exhaustive RF hyperparameter search but a strong, reproducible classical baseline. No `StandardScaler` is required because tree decisions depend on feature ordering and thresholds rather than feature magnitude.

How Random Forest makes a classification

Each decision tree is fitted to a bootstrap sample of the labelled pixels. At every node, only a random subset of predictors is considered, and the split that produces the purest child nodes is selected. With scikit-learn defaults, node impurity is measured using the Gini criterion. For a node t containing K classes, with $p(k|t)$ denoting the proportion of samples from class k , the impurity is:

$$G(t) = 1 - \sum_{k=1}^K p(k|t)^2 \quad (1)$$

A candidate split is preferred when it gives the largest weighted reduction in impurity. If N_t samples reach the parent node and N_L and N_R reach the two children, the decrease is:

$$\Delta G = G(t) - (N_L/N_t)G(t_L) - (N_R/N_t)G(t_R) \quad (2)$$

Repeating this process creates many decorrelated trees. For a new pixel x , every tree $h_b(x)$ casts one class vote and the forest returns the class with the largest vote share. For B trees:

$$\hat{y}(x) = \operatorname{argmax}_k [(1/B) \sum_{b=1}^B I(h_b(x)=k)] \quad (3)$$

RF feature-set comparison

Feature set	Mean CV accuracy	Std. accuracy	Mean Macro F1	Std. F1	CV time (s)
Raw bands	0.996917	0.000715	0.996918	0.000713	11.34
Raw bands + indices	0.996917	0.000841	0.996918	0.000841	9.63

Table 7a. Random Forest five-fold cross-validation results on the 7,136-sample development set.

The extended nine-feature representation was retained because it ranked first by mean Macro F1, although its advantage over the six raw bands was numerically negligible (approximately 0.00000016 Macro F1). The experiment therefore indicates that the RF already extracts nearly all useful separation from the raw bands, while the index-augmented representation remains the selected production input under the fixed ranking rule.

Production use

After the final hold-out comparison selected RF as the winning classifier, a new production model was trained on all 8,920 balanced labelled pixels rather than only the 7,136 development subset. The model then classified the complete scene in chunks of 1,000,000 pixels to control memory use. Predictions were written to a single-band uint8 GeoTIFF with `NoData=0` while preserving the reference geotransform, CRS and dimensions.

Selected RF configuration
200 trees <code>random_state=42</code> six raw bands + NDVI/NDWI/NDBI no feature scaling final production training set: 8,920 balanced pixels

5.6 Multi-Layer Perceptron Method

The Multi-Layer Perceptron is a fully connected feed-forward neural network trained by back-propagation. Unlike Random Forest, its optimization is sensitive to predictor scale. Each MLP experiment is therefore wrapped in a scikit-learn Pipeline containing StandardScaler followed by MLPClassifier. During cross-validation the scaler is fitted only on the training fold, preventing information from the validation fold from influencing preprocessing.

Six architectures were tested with both feature sets. The experiments vary hidden-layer depth/width, activation function and initial learning rate. ReLU configurations examine one- and two-layer networks, while tanh configurations provide a contrasting smooth bounded activation. The configuration names are retained throughout the notebook so cross-validation, final testing, plotting and export refer to the same selected model.

How the MLP learns the class boundaries

The MLP receives one feature vector per pixel. StandardScaler is fitted inside each training fold only, so feature j is standardized using training-fold statistics:

$$x'_j = (x_j - \mu_j) / \sigma_j \quad (4)$$

A fully connected layer transforms the previous activations by a weighted sum plus bias and then applies a nonlinear activation:

$$a^{(l)} = f(W^{(l)}a^{(l-1)} + b^{(l)}) \quad (5)$$

The tested activations are ReLU, $f(z)=\max(0,z)$, and tanh. In the selected MLP_4, nine inputs feed 100 hidden units, then 50 hidden units, and four class outputs. Softmax converts output scores to probabilities:

$$p_k = \exp(o_k) / \sum_j \exp(o_j) \quad (6)$$

Training minimizes multiclass cross-entropy, with gradients obtained by back-propagation. MLPClassifier uses Adam by default; max_iter=300, early_stopping=True, validation_fraction=0.10 and n_iter_no_change=15 are fixed in the notebook, while the initial learning rate is tuned (Rumelhart et al., 1986; Kingma and Ba, 2015).

$$L = -(1/N) \sum_i \sum_k y_{ik} \log(p_{ik}) \quad (7)$$

Configuration	Hidden layers	Activation	Initial learning rate
MLP_1	(50,)	ReLU	0.001
MLP_2	(100,)	ReLU	0.001
MLP_3	(50, 25)	ReLU	0.001
MLP_4	(100, 50)	ReLU	0.0005
MLP_5	(50,)	tanh	0.001
MLP_6	(100,)	tanh	0.0005

Table 4. MLP hyperparameter configurations evaluated with both feature sets.

The best development-set MLP result was MLP_4 with raw bands plus indices: mean CV accuracy 0.9950 and mean Macro F1 0.9950. This architecture has two hidden layers with 100 and 50 units, ReLU activation and an initial learning rate of 0.0005. MLP_3 was a close second with the same feature set, while the wider experiment still shows greater architecture sensitivity than RF because the six tested networks span a substantially broader score range.

Operational implication

The MLP provides a useful comparison because it tests whether a modest neural network can exploit the same feature vectors without spatial convolution. However, it introduces additional preprocessing, convergence behaviour and tuning choices. The final selection therefore considers not only model complexity but also whether those additional requirements translate into better independent test performance.

5.7 Validation Metrics and Reproducibility Controls

Evaluation metrics

Overall accuracy is the fraction of test samples classified correctly. Precision measures how reliable the predictions assigned to a class are, while recall measures how much of the true class is recovered. F1 is the harmonic mean of precision and recall. Macro F1 averages the four class-level F1 scores with equal weight, making it especially suitable for the balanced evaluation set. Cohen's kappa provides an additional chance-corrected agreement statistic. Confusion matrices are reported with true classes in rows and predicted classes in columns.

Cross-validation

Five-fold StratifiedKfold with shuffling and random_state=42 is used on the 7,136-sample development subset. Stratification preserves the class proportions in each fold. Mean scores summarize expected development performance, while standard deviation shows fold-to-fold sensitivity. Training time is recorded for comparison but is treated as secondary to predictive quality because runtime can vary with hardware and convergence.

Reproducibility controls

- Centralized project paths and explicit output directories.
- Automatic verification that required inputs exist before modelling.
- Fixed class-value map: 1 Urban, 2 Vegetation, 3 Water, 4 Bare Soil.
- Fixed random seed for sampling, split, CV and model initialization where applicable.
- Identical train/test indices for the raw and extended feature matrices.
- Scaler included inside the MLP cross-validation Pipeline to prevent leakage.
- Saved CSV tables for training quality, CV results, feature comparison, final models, map metadata and area statistics.
- Saved PNG figures and georeferenced GeoTIFF outputs with verification in the final notebook step.

Methodological caution

The hold-out subset is independent of tuning decisions but is not geographically independent because test pixels still come from the same collection of training polygons. Spatial autocorrelation among nearby pixels may therefore make the absolute accuracy optimistic compared with validation on a separate tile or separately digitized polygons. This limitation is stated explicitly in the discussion rather than hidden by the high scores.

Validation and Reproducibility Controls

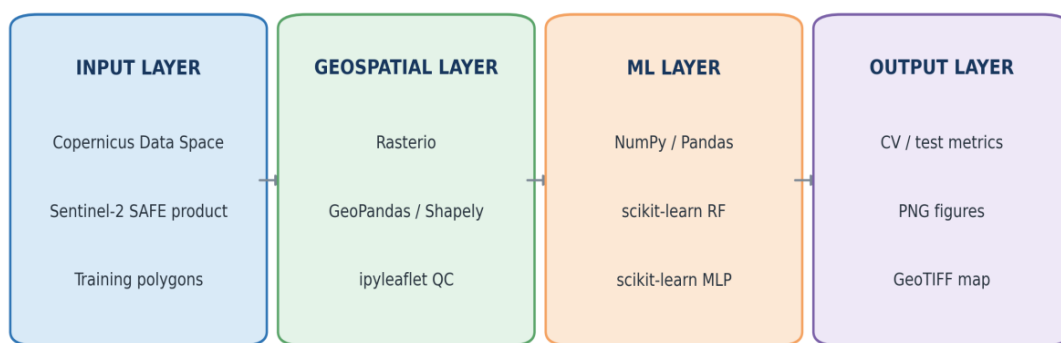


6. Software Implementation Architecture

The implementation is designed as a sequential Jupyter Notebook with 27 numbered processing steps. Each step performs one coherent task, checks its prerequisites, prints interpretable status information and writes reusable outputs where appropriate. This structure supports both academic review and practical reuse: a user can understand where data acquisition ends, where training labels are created, where modelling begins, and which files form the final result package.

Python provides the common execution environment. Rasterio handles raster reading, metadata, reprojection, masking and GeoTIFF writing. GeoPandas and Shapely manage vector geometry and CRS transformations. ipyleaflet and ipywidgets provide the interactive training interface. NumPy and Pandas manage feature arrays and tabular results. scikit-learn implements Random Forest, MLP, scaling, cross-validation, splitting and metrics. Matplotlib creates the final figures. Pillow is used to prepare image overlays for interactive display.

Software and Data Architecture



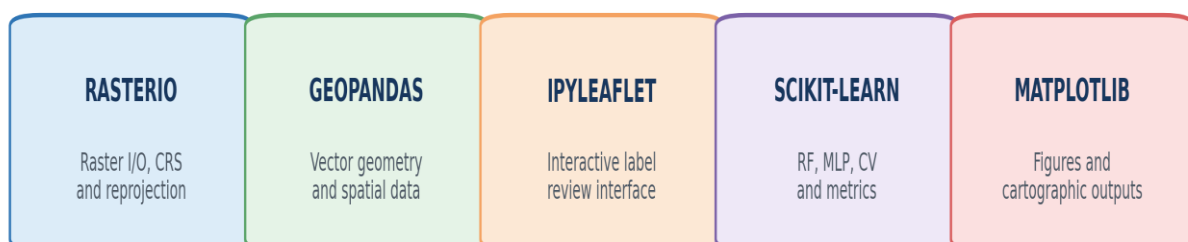
Modular separation of data access, geospatial processing, model selection and production outputs.

Figure 3 (reproduced). The architecture separates input, geospatial processing, machine-learning and output responsibilities.

Implementation principles

- Fail early when a required input or CRS is missing.
- Keep scientific arrays and visual display layers conceptually separate.
- Use saved intermediate products to reduce unnecessary repeated preprocessing.
- Make model selection dynamic: later steps read the actual winning classifier and feature set instead of hard-coding a choice.
- Preserve raster geometry and class values in production outputs.
- Organize all final tables, figures and GeoTIFFs into one verified Results directory.

Implementation Stack



6.1 Notebook Workflow: Steps 1-14

Step	Notebook stage	Purpose / main output
1	Project configuration	Defines directories, classes, switches and shared settings.
2	Sentinel-2 acquisition utilities	Implements CDSE search/download and preparation helpers.
3	Area / tile selection	Defines or resolves the target Sentinel-2 tile/area.
4	Search, select, download, prepare	Retrieves Level-2A data and prepares six required bands.
5	High-resolution RGB preview	Creates a stretched RGB image for visual inspection.
6	Interactive training polygons	Loads/edits/draws polygons; checks balance and labels.
7	Connect prepared project paths	Links prepared rasters and output locations.
8	Check all input files	Confirms required raster/vector inputs exist.
9	Load Sentinel-2 bands	Reads the six aligned raster bands.
10	Stack Sentinel-2 bands	Builds the multi-band raster cube.
11	NoData / valid-pixel mask	Excludes invalid pixels before modelling.
12	Compute indices	Calculates NDVI, NDWI and NDBI.
13	Load and validate polygons	Checks class fields, CRS and geometry.
14	Rasterize labels / extract pixels	Associates training polygons with raster feature vectors.

Table 5. Notebook workflow, steps 1-14.

These early stages encapsulate most geospatial data engineering. Their outputs are deterministic, inspectable arrays and reference labels that can be consumed by the machine-learning stages without repeating acquisition or geometry operations.

6.2 Notebook Workflow: Steps 15-27

Step	Notebook stage	Purpose / main output
15	Inspect sample distribution	Reports pre-balance pixel and polygon counts.
16	Balance training pixels	Selects 2,230 pixels per class; builds 8,920-sample datasets.
17	Stratified train/test split	Creates 7,136 development and 1,784 hold-out samples.
18	RF five-fold CV	Compares raw bands and raw bands plus indices.
19	MLP experiments	Runs six architectures with both feature sets.
20	Export CV results	Combines and ranks model/feature experiments.
21	Final training and hold-out evaluation	Fits CV-selected RF and MLP; computes final metrics.
22	Confusion matrices	Plots and saves class-level error matrices.
23	Full-scene classification	Retrains winning model on all balanced labels and predicts scene.
24	Final model table	Exports consistent RF/MLP comparison metrics.
25	Comparison figures	Creates hold-out and feature-engineering plots.
26	Final land-cover map	Validates class values and exports 300-DPI map.
27	Organize and verify outputs	Checks essential deliverables and result inventory.

Table 6. Notebook workflow, steps 15-27.

A separate post-processing cell can be inserted immediately after full-scene classification to create `RF_landcover_sieved_display.tif` for map presentation. This derivative file is intentionally outside the evaluation chain; the raw Step 23 raster remains the source for scientific statistics.

6.3 User Manual and Deliverables

Recommended execution sequence

1. Open the notebook and set PROJECT_DIR and other user settings in Step 1.
2. Run the Sentinel-2 acquisition/preparation cells or connect to an already prepared Level-2A product.
3. Use Step 6 to review existing training polygons or create new ones. Confirm each class reaches the polygon/pixel targets and run Check Labels.
4. Run Steps 7-17 to load the data, calculate features, extract labels, balance samples and create the fixed 80/20 split.
5. Run Steps 18-20 to perform RF and MLP cross-validation. Do not inspect the hold-out test during tuning.
6. Run Steps 21-22 once for final model evaluation and confusion matrices.
7. Run Step 23 to train the selected production model on all balanced labelled pixels and classify the complete scene.
8. Optionally create the 8-pixel sieve display raster for visual presentation only.
9. Run Steps 24-27 to export comparison tables, figures, final map and the verified Results package.

Core deliverables

File	Purpose
training_polygon_quality.csv	Final reference-polygon and pixel quality by class.
cross_validation_results.csv	Detailed RF/MLP development experiments.
final_model_comparison.csv	Selected RF and MLP CV/hold-out summary.
feature_engineering_summary.csv	Raw versus raw+indices Macro F1 comparison.
rf_confusion_matrix.png / mlp_confusion_matrix.png	Class-level test errors.
landcover_classification_best_model.tif	Scientific raw production classification.
landcover_area_summary.csv	Raw class pixel counts, areas and percentages.
landcover_map.png	Final raw land-cover figure.
RF_landcover_sieved_display.tif / .png	Optional cleaned cartographic display.
final_map_model_info.csv	Model and feature set used for production mapping.

Re-run expectations

If the same input product, final reviewed training polygons, software versions and random seed are used, the workflow is designed to reproduce the same split logic, model ranking and exported structure. Small runtime differences are expected across computers, but predictive metrics should remain stable when the numerical environment is equivalent.

7. Results: Training Quality and Cross-Validation Overview

The final reference set achieved equal spatial sampling effort with 40 polygons for every class. Pixel availability remained moderately unequal because polygon footprints differ, but all classes exceeded the 2,000-pixel target. Urban was the smallest class with 2,230 valid pixels and therefore set the balancing threshold. After balancing, all four classes contributed exactly 25% of the 8,920-sample dataset. Figure 6b illustrates the interactive QC review state; final counts are reported in Table 2.

Random Forest development results

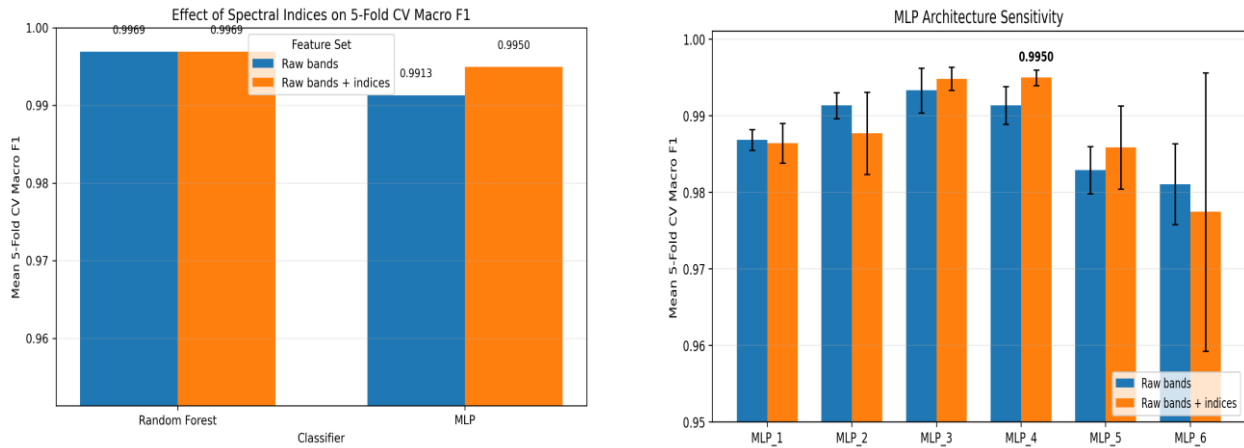
Feature set	CV Accuracy	CV Macro F1	Std. F1	Interpretation
Raw bands	0.996917	0.996918	0.000713	Effectively equivalent baseline
Raw bands + indices	0.996917	0.996918	0.000841	Selected - negligible numerical edge

Table 7. RF development-set feature comparison.

Random Forest achieved extremely high and stable development performance. With raw bands alone, mean five-fold Macro F1 was 0.996918. Adding NDVI, NDWI and NDBI produced 0.996918 with a standard deviation of 0.000841. The difference between the two means is only about 0.0000002, so the indices did not materially improve RF performance; the extended feature set was retained because it held the top numerical rank under the fixed comparison protocol.

This result establishes an important baseline for the rest of the report: engineered spectral indices were evaluated rather than assumed to help. For RF, the raw and extended feature sets were effectively equivalent, showing that the tree ensemble could already recover the dominant class contrasts from the six original bands.

7.1 Feature Engineering and MLP Sensitivity



(a) Effect of spectral indices on selected model families. (b) Macro F1 sensitivity across six MLP architectures.

Figures 7-8. Feature-engineering and MLP sensitivity results.

The spectral-index experiment produced different effects across model families. Random Forest mean CV Macro F1 was 0.996918 with raw bands and 0.996918 with indices, a practically zero change of about +0.0000002. For the selected MLP_4 architecture, Macro F1 increased from 0.991309 to 0.994957, an improvement of 0.003647. The deterministic index transformations therefore helped the selected MLP more clearly, while providing no meaningful gain to RF.

Configuration	Hidden layers	Activation / LR	Raw CV F1 +/- SD	Raw + indices CV F1 +/- SD	Best feature set
MLP_1	(50,)	relu / 0.001	0.9868 +/- 0.0014	0.9864 +/- 0.0026	Raw bands
MLP_2	(100,)	relu / 0.001	0.9913 +/- 0.0017	0.9877 +/- 0.0054	Raw bands
MLP_3	(50, 25)	relu / 0.001	0.9933 +/- 0.0029	0.9948 +/- 0.0015	Raw + indices
MLP_4	(100, 50)	relu / 0.0005	0.9913 +/- 0.0025	0.9950 +/- 0.0010	Raw + indices
MLP_5	(50,)	tanh / 0.001	0.9829 +/- 0.0031	0.9858 +/- 0.0054	Raw + indices
MLP_6	(100,)	tanh / 0.0005	0.9810 +/- 0.0053	0.9774 +/- 0.0182	Raw bands

Note. Yellow highlights the best MLP architecture; green highlights its winning cross-validation score. Random Forest CV results are reported separately in Table 7a. Full run-level results remain available in `cross_validation_results.csv`.

Table 7b. MLP five-fold cross-validation results across architectures and feature sets.

Table 7b summarizes all six MLP architectures under both feature sets. MLP_4, using hidden layers (100, 50), ReLU and learning rate 0.0005, was the strongest neural-network configuration with raw bands plus indices, reaching 0.9950 +/- 0.0010 mean CV Macro F1. MLP_3 with indices followed closely at 0.9948 +/- 0.0015. Table 7a shows that Random Forest remained slightly stronger and more stable overall at 0.9969 +/- 0.0008.

Cross-Validation Takeaways

RF BEST 0.9969 +/- 0.0008 Macro F1 Raw + indices	MLP BEST 0.9950 +/- 0.0010 Macro F1 MLP_4 + indices	FEATURE EFFECT RF: essentially unchanged MLP: +0.00365 Macro F1
--	---	--

7.2 Final Hold-Out Evaluation

After cross-validation selected RF with raw bands plus indices and MLP_4 with raw bands plus indices, both models were retrained on the entire 7,136-sample development subset and evaluated once on the untouched 1,784-pixel test subset. The test set contains exactly 446 observations per class, so overall accuracy and Macro F1 can be compared directly without class-prevalence effects.

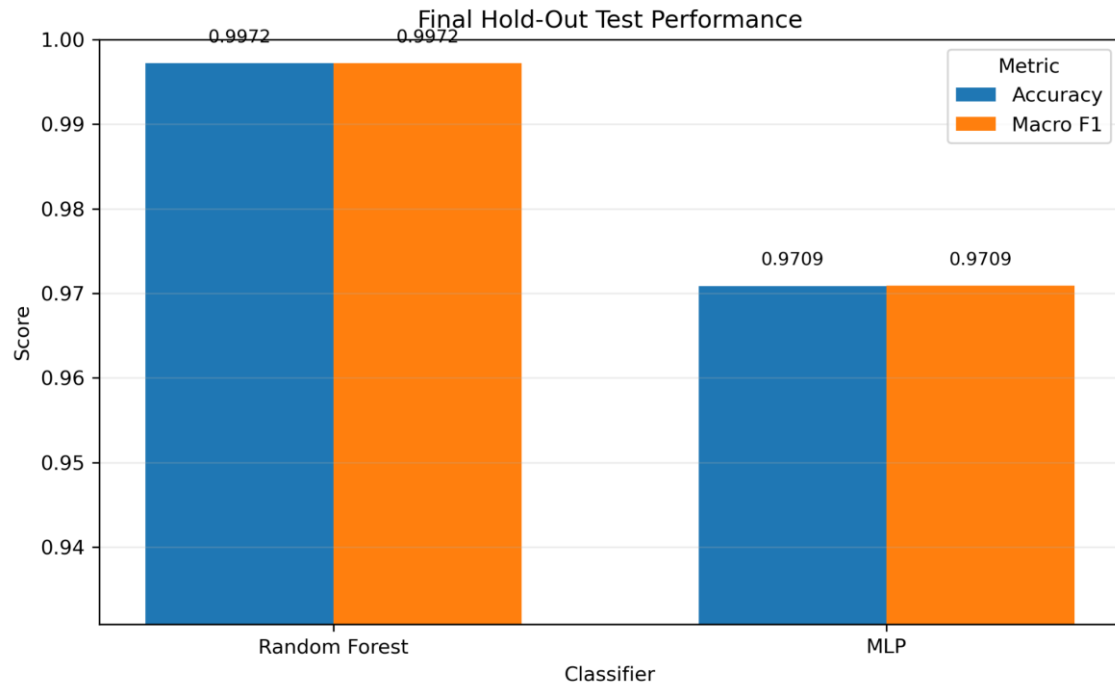


Figure 9. Final hold-out accuracy and Macro F1 for the CV-selected Random Forest and MLP configurations.

Metric	Random Forest	Selected MLP
Best feature set	Raw bands + indices	Raw bands + indices
Mean CV accuracy	0.996917 +/- 0.000841	0.994955 +/- 0.001031
Mean CV Macro F1	0.996918 +/- 0.000841	0.994957 +/- 0.001031
Hold-out accuracy	0.997197	0.970852
Hold-out Macro precision	0.997199	0.970921
Hold-out Macro recall	0.997197	0.970852
Hold-out Macro F1	0.997197	0.970882
Cohen's kappa	0.996263	0.961136
Final fit + test prediction time	1.56 s	2.26 s

Table 8. Final hold-out model comparison.

Random Forest was clearly the best final model. RF achieved 0.9972 hold-out accuracy and 0.9972 Macro F1, compared with 0.9709 accuracy and 0.9709 Macro F1 for the CV-selected MLP_4. Development performance had been much closer (0.9969 versus 0.9950 Macro F1), so the larger hold-out gap is an important result: the MLP configuration that ranked best in cross-validation generalized less strongly to the untouched test pixels. RF also required no feature scaling and had the shorter final fit-plus-test prediction time in this run.

7.3 Confusion-Matrix Analysis

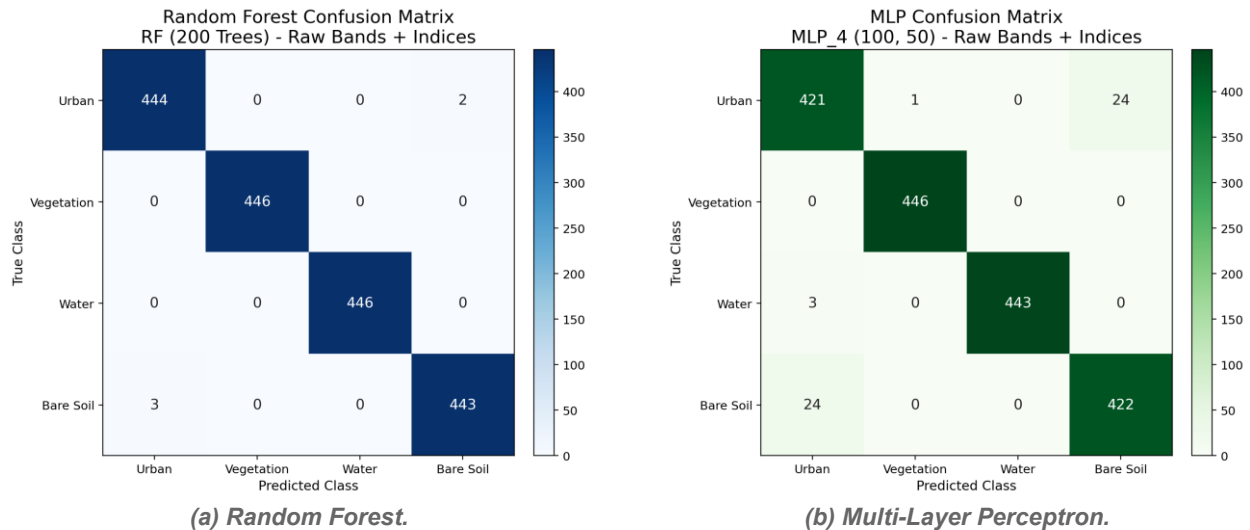


Figure 10. Final hold-out confusion matrices. Rows are true classes and columns are predicted classes.

Random Forest per-class metrics

Class	Precision	Recall	F1	Support
Urban	0.9933	0.9955	0.9944	446
Vegetation	1.0000	1.0000	1.0000	446
Water	1.0000	1.0000	1.0000	446
Bare Soil	0.9955	0.9933	0.9944	446

MLP per-class metrics

Class	Precision	Recall	F1	Support
Urban	0.9397	0.9439	0.9418	446
Vegetation	0.9978	1.0000	0.9989	446
Water	1.0000	0.9933	0.9966	446
Bare Soil	0.9462	0.9462	0.9462	446

The RF confusion matrix is nearly diagonal: Vegetation and Water achieve perfect recall, while the only errors are two Urban pixels predicted as Bare Soil and three Bare-Soil pixels predicted as Urban. The MLP matrix shows a different pattern, with most errors concentrated in Urban/Bare-Soil confusion (24 pixels in each direction), plus one Urban pixel predicted as Vegetation and three Water pixels predicted as Urban. These matrices explain the stronger RF hold-out score and show that MLP's main generalization weakness is the spectral boundary between built and exposed-soil surfaces.

7.4 Full-Scene Classification and Area Statistics

The final production Random Forest was trained on all 8,920 balanced labelled samples using the six raw bands plus NDVI, NDWI and NDBI and applied to every valid pixel in the scene. Prediction was performed in one-million-pixel chunks to limit memory use. The classified raster preserves the original 10 m grid and EPSG:32632 georeferencing and contains only class values 1-4. Coverage is 100% because all 120,560,400 raster pixels passed the final valid-pixel mask.

Final Land-Cover Classification
Random Forest — Raw bands + indices

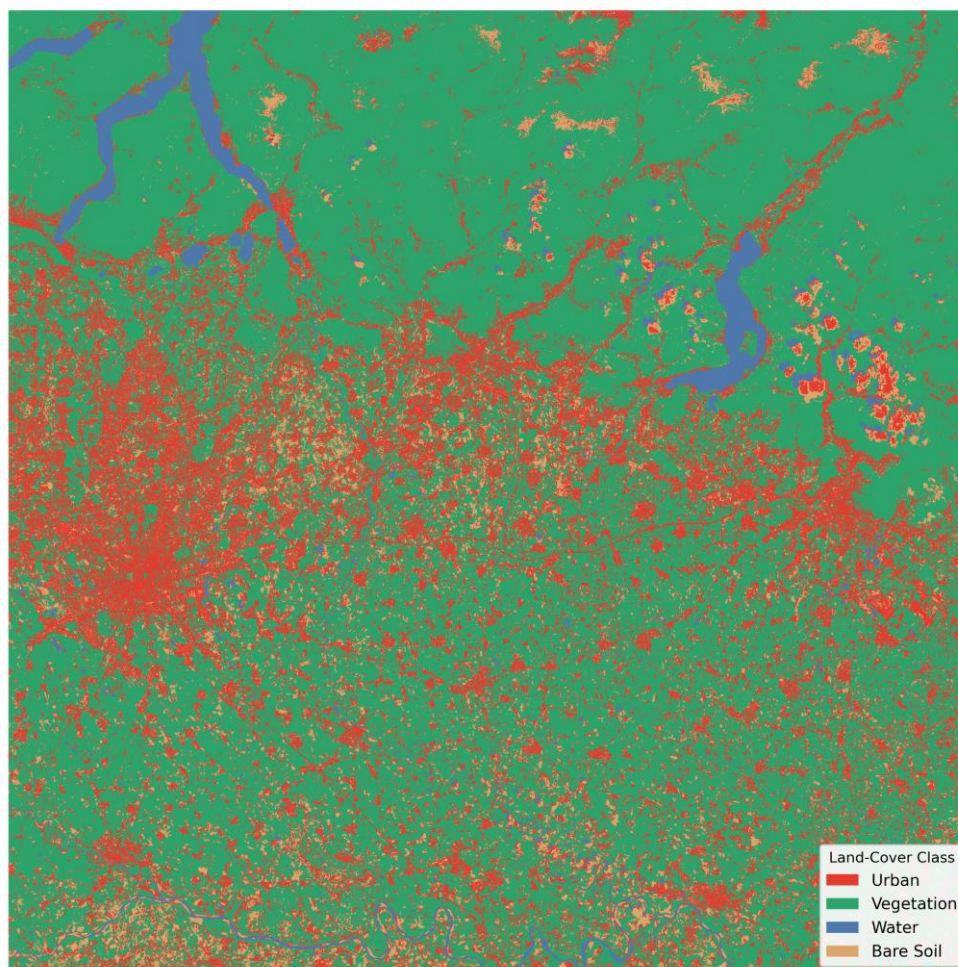


Figure 11. Final raw Random Forest land-cover classification generated from the six Sentinel-2 bands plus NDVI, NDWI and NDBI.

Class	Pixel count	Area (km ²)	Percentage
Urban	26,692,487	2,669.25	22.14%
Vegetation	82,688,766	8,268.88	68.59%
Water	3,286,667	328.67	2.73%
Bare Soil	7,892,480	789.25	6.55%
Total	120,560,400	12,056.04	100.00%

Table 10. Area statistics calculated from the raw scientific classification.

Vegetation is the dominant mapped class at 68.59%, followed by Urban at 22.14%, Bare Soil at 6.55% and Water at 2.73%. The broad Vegetation category includes forest, grassland, crops and other green surfaces; it should not be interpreted as agricultural land alone. The revised distribution is consistent with the final training review and the spatial pattern visible in the production map.

7.5 Cartographic Post-Processing

Pixel-based classification often produces a salt-and-pepper appearance because each 10 m pixel is predicted independently. Small isolated regions can therefore be visually distracting even when the underlying classifier is accurate. To improve map readability, an 8-pixel sieve filter was applied after production classification. The filter removes small connected patches below the selected threshold by merging them into surrounding regions.

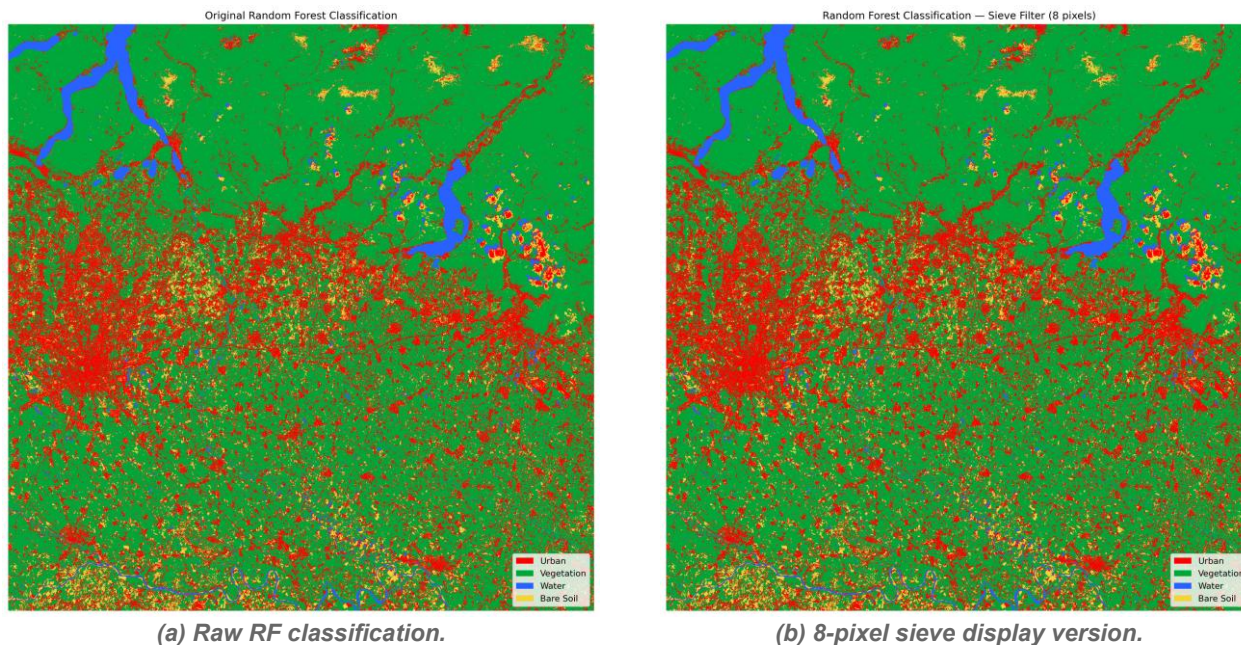


Figure 12. Raw scientific classification and the cartographic 8-pixel sieve version.

The sieve modified 1,492,682 of 120,560,400 valid pixels, corresponding to 1.238% of the scene. The resulting display proportions were 22.264% Urban, 68.766% Vegetation, 2.719% Water and 6.251% Bare Soil. These values are shown only to quantify the filter effect; the report uses the raw classification for all scientific area statistics and evaluation claims.

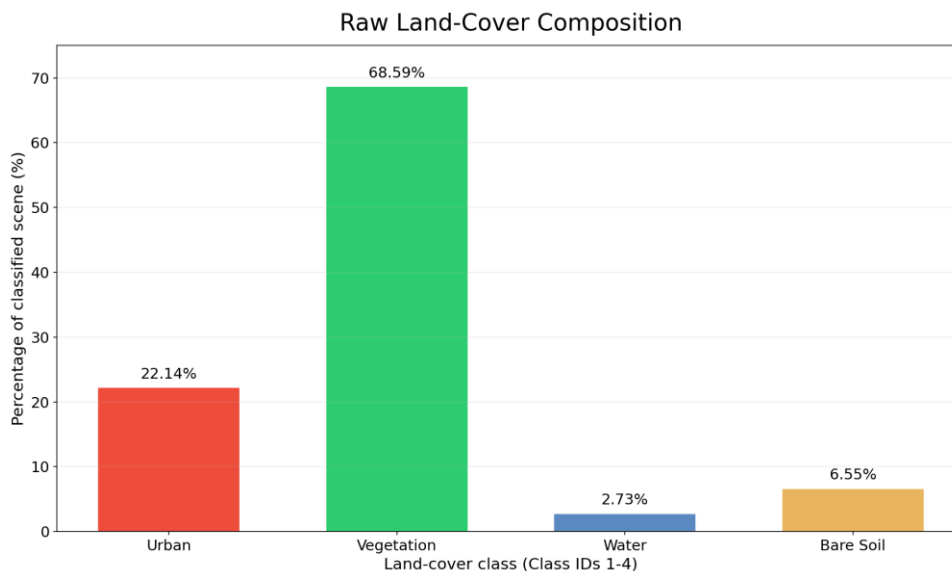


Figure 13. Land-cover composition calculated from the raw production raster, shown in class-ID order: Urban, Vegetation, Water and Bare Soil.

Interpretation rule

Accuracy, confusion matrices, kappa and area statistics = raw classification. Sieve output = cartographic presentation only. No accuracy result is recalculated after filtering.

8. Discussion and Limitations

The results support Random Forest as the preferred model for this four-class Sentinel-2 problem. RF achieved 99.72% hold-out accuracy with 0.9969 mean development Macro F1 and very small fold-to-fold variation. The selected MLP_4 reached a similarly strong 0.9950 development Macro F1 but only 97.09% hold-out accuracy. The final choice therefore reflects both RF's slightly stronger and more stable cross-validation result and its substantially better performance on the untouched test subset.

The feature-engineering experiment is also informative. NDVI, NDWI and NDBI had essentially no effect on Random Forest: the raw and extended RF Macro F1 values differ by only about 0.00000016. In contrast, the selected MLP_4 improved from 0.991309 with raw bands to 0.994957 with indices (+0.003647). Because the indices are deterministic functions of the source bands, this suggests that explicit spectral ratios simplified useful decision boundaries for the MLP even though they added almost no value to the tree ensemble.

The confusion matrices show that the remaining RF errors are exceptionally sparse and localized, with only five mistakes across the 1,784 test pixels and all of them involving Urban versus Bare Soil. MLP errors are also concentrated in the Urban/Bare-Soil boundary, but at a much higher frequency, which accounts for its lower hold-out score. The final map retains fine-scale pixel variability, as expected from a pixel-based workflow, and this motivates the separate sieve display without altering the scientific classification.

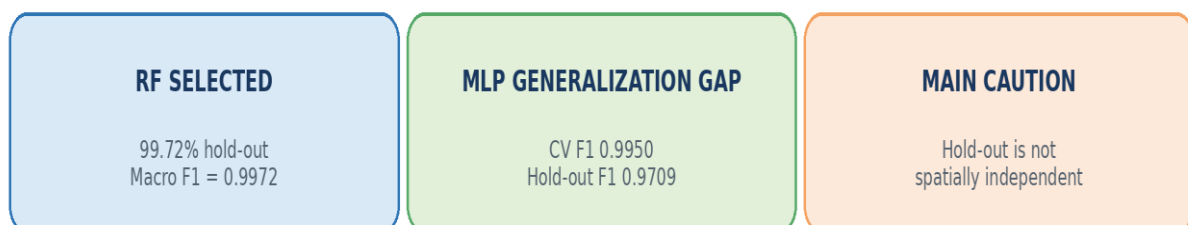
Limitations

- Training and hold-out pixels originate from the same reference-polygon collection. Spatial correlation can make the reported accuracy optimistic compared with geographically independent validation.
- The analysis uses a single summer Sentinel-2 acquisition. Seasonal crop cycles, leaf-off conditions, snow and multi-temporal behaviour are not represented.
- The thematic legend contains only four broad classes. Vegetation combines multiple ecological and agricultural sub-types, while Bare Soil includes several dry/exposed surface conditions.
- Balanced sampling is appropriate for model comparison but does not reproduce the natural frequency of classes across the complete tile.
- The RF parameter search is intentionally light, and the MLP search covers six architectures rather than an exhaustive optimization space.
- The 8-pixel sieve is a cartographic choice and can alter local patch boundaries; it must not be interpreted as improved classifier accuracy.

Future improvements

Future work could use spatially independent validation polygons, multi-season Sentinel-2 composites, additional land-cover subclasses, calibrated probability maps, spatial-context models such as CNN-based classifiers, or object-based post-processing. These extensions would test whether the already high pixel-based accuracy transfers to more detailed and geographically independent conditions.

Final Interpretation



9. Conclusion

This project implemented a complete Sentinel-2 land-cover classification workflow from data acquisition to final georeferenced product. The pipeline harmonized six multispectral bands, created and quality-controlled interactive training polygons using all-class spectral checks, extracted and balanced labelled pixels, compared raw spectral bands with three engineered indices, evaluated Random Forest and six MLP architectures under leakage-free five-fold cross-validation, preserved an untouched hold-out test, and classified the complete 120.56-million-pixel scene.

Random Forest with 200 trees and the six raw bands plus NDVI, NDWI and NDBI was the best configuration. It achieved 0.9969 $\pm$ 0.0008 mean development Macro F1, 0.9972 hold-out accuracy, 0.9972 hold-out Macro F1 and 0.9963 Cohen's kappa. The selected MLP_4 reached 0.9950 $\pm$ 0.0010 development Macro F1 but 0.9709 hold-out accuracy and 0.9709 Macro F1. The production RF mapped 22.14% Urban, 68.59% Vegetation, 2.73% Water and 6.55% Bare Soil across the 12,056 km² raster footprint.

The final workflow is technically justified by both predictive evidence and software-engineering controls. It is reproducible, preserves geospatial metadata, reports class-level errors, uses all-class advisory label QC, and keeps scientific results separate from an optional 8-pixel sieve used only to improve cartographic readability. Within the project scope, the results show that careful reference-data review and controlled validation can allow a classical ensemble and a compact neural network to achieve very strong Sentinel-2 land-cover performance, with Random Forest retaining the best overall balance of accuracy, stability and operational simplicity.

References

- Belgiu, M., & Dragut, L. (2016). Random forest in remote sensing: A review of applications and future directions. *ISPRS Journal of Photogrammetry and Remote Sensing*, 114, 24-31.
- Breiman, L. (2001). Random forests. *Machine Learning*, 45, 5-32.
- Congalton, R. G. (1991). A review of assessing the accuracy of classifications of remotely sensed data. *Remote Sensing of Environment*, 37(1), 35-46.
- Drusch, M., Del Bello, U., Carlier, S., et al. (2012). Sentinel-2: ESA's optical high-resolution mission for GMES operational services. *Remote Sensing of Environment*, 120, 25-36.
- Foody, G. M. (2002). Status of land cover classification accuracy assessment. *Remote Sensing of Environment*, 80(1), 185-201.
- Maxwell, A. E., Warner, T. A., & Fang, F. (2018). Implementation of machine-learning classification in remote sensing: An applied review. *International Journal of Remote Sensing*, 39(9), 2784-2817.
- Kingma, D. P., & Ba, J. (2015). Adam: A method for stochastic optimization. *International Conference on Learning Representations (ICLR)*.
- McFeeters, S. K. (1996). The use of the Normalized Difference Water Index (NDWI) in the delineation of open water features. *International Journal of Remote Sensing*, 17(7), 1425-1432.
- Pedregosa, F., Varoquaux, G., Gramfort, A., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825-2830.
- Rumelhart, D. E., Hinton, G. E., & Williams, R. J. (1986). Learning representations by back-propagating errors. *Nature*, 323, 533-536.
- Rouse, J. W., Haas, R. H., Schell, J. A., & Deering, D. W. (1974). Monitoring vegetation systems in the Great Plains with ERTS. *Third Earth Resources Technology Satellite-1 Symposium*, NASA, 309-317.
- Zha, Y., Gao, J., & Ni, S. (2003). Use of normalized difference built-up index in automatically mapping urban areas from TM imagery. *International Journal of Remote Sensing*, 24(3), 583-594.
- European Space Agency. Sentinel-2 Level-2A product documentation and user guidance.
- Politecnico di Milano. Geoinformatics Project (GIP) 2025/2026 project procedures and evaluation guidance.

Project Repository

The complete Jupyter Notebook, source code, training data and project outputs are available at:
<https://github.com/Tarigalmaghrabi/sentinel2-landcover-rf-mlp>